



9563 - Don't Let the Dust Settle: Observing Planetesimal Collisions in the Beta Pictoris Debris Disk

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Arin Avsar (PI)	University of Arizona
Dr. Kevin Wagner (CoI) (CoPI)	University of Arizona
Prof. Daniel Apai (CoI)	University of Arizona
Prof. Mark Wyatt (CoI) (ESA Member)	University of Cambridge
Dr. Chris Stark (CoI)	NASA Goddard Space Flight Center
Isabel Rebolledo (CoI) (ESA Member)	ESA, European Space Astronomy Centre

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Beta Pic - SW Roll 1	NIRCam Coronagraphic Imaging	(1) -bet-Pic
	2	Beta Pic - SW Roll 2	NIRCam Coronagraphic Imaging	(1) -bet-Pic
	3	Alpha Pic - SW PSF	NIRCam Coronagraphic Imaging	(2) -alf-Pic

ABSTRACT

Beta Pictoris is a dynamically active system that has undergone major collisions in the recent past. In this program, we seek to use multi-epoch imaging to uncover collisional remnants in the Beta Pictoris disk from a population of planetesimals that was not accessible in Beta Pic before JWST. Simulations of multi-epoch imaging of Beta Pic with NIRCam shows it is capable of detecting remnants down to 0.1 Ceres masses. This is a 10–100 fold increase in the number of planetesimals NIRCam can detect collisions between compared to HST. This translates to a 62% chance of a detectable collision with NIRCam occurring in Beta Pic since the first epoch in 2022, making Beta Pic an ideal target for this study. We propose a

second epoch of NIRCам imaging of Beta Pic to search for collision events between 2022 and 2026. Multi-epoch imaging, even in the case of non-detections, can provide valuable constraints on the underlying planetesimal population. Cycle 5 is the ideal time to perform these observations. Since debris clouds are brightest and most detectable 2–4 years after collision, performing observations 4 years apart allows for the detection of low-mass collisions while maximizing the time between observations allows for a high probability of a collision event. We highlight the urgency in starting regular monitoring this cycle as low observing cadences can result in $\sim 50\%$ of collision events going undetected due to cloud dissipation. Ultimately, this program seeks to lay the groundwork for characterizing the formation and evolution of extrasolar planetesimal belts through the lens of major collisions.

OBSERVING DESCRIPTION

These observations aim to obtain a second epoch of data on the Beta Pictoris debris disk. These observations will follow that of GTO 1411 NIRCам observations in the F182M and F250M filters, using the M210R mask. We will observe in near identical observing modes to minimize systematics in our observations in order to maximize our sensitivity to astrophysical changes in the disk. The only difference over the GTO 1411 observations will be SW+LW coronagraphy, which was not available in Cycle 1. Our datasets will be contemporaneous, and the exposures are linked. As a result, the F182M observations will be identical to the GTO 1411 observations, while the F250M will have a different readout pattern and slightly longer exposure time compared to GTO 1411 observations. Additionally, the observing mode for the F182M filter has already demonstrated its capability to detect the disk with high SNR. We expect to be sensitive to the remnants of colliding planetesimals down to 0.1 Ceres masses per colliding progenitor when combined with the first epoch of GTO observations.

Proposal 9563 - Targets - Don't Let the Dust Settle: Observing Planetsimal Collisions in the Beta Pictoris Debris Disk

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	-bet-Pic	RA: 05 47 17.0877 (86.8211988d) Dec: -51 03 59.44 (-51.06651d) Equinox: J2000	Proper Motion RA: 5.16 mas/yr Proper Motion Dec: 84.041 mas/yr Parallax: 0.0509307" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Coronagraphic]				
	(2)	-alf-Pic	RA: 06 48 11.4551 (102.0477296d) Dec: -61 56 29.00 (-61.94139d) Equinox: J2000	Proper Motion RA: -66.07 mas/yr Proper Motion Dec: 242.97 mas/yr Parallax: 0.03378000000000004" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Coronagraphic]				

Proposal 9563 - Observation 1 - Don't Let the Dust Settle: Observing Planetesimal Collisions in the Beta Pictoris Debris Disk

Observation	Proposal 9563, Observation 1: Beta Pic - SW Roll 1										Fri Apr 03 16:00:09 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Coronagraphic Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Beta Pic - SW Roll 1 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	-bet-Pic	RA: 05 47 17.0877 (86.8211988d) Dec: -51 03 59.44 (-51.06651d) Equinox: J2000			Proper Motion RA: 5.16 mas/yr Proper Motion Dec: 84.041 mas/yr Parallax: 0.0509307" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Calibration Description=[Coronagraphic]										
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	F210M	BRIGHT (ND Square)	RAPID	9	1	1	1.825	25161	
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern		
	A		MASK210R		false		SUB640A210R		NONE		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F182M	F250M	RAPID	4	90	1	90	1885.471		
PSF References	Alpha Pic - SW PSF (Obs 3) (PSF Reference; Filters [F182M/F250M]) Additional Justification: false										

Proposal 9563 - Observation 1 - Don't Let the Dust Settle: Observing Planetesimal Collisions in the Beta Pictoris Debris Disk

Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible Aperture PA Offset 2 from 1 by 10 to 14 Degrees (Same offsets in V3)
----------------------	--

Proposal 9563 - Observation 2 - Don't Let the Dust Settle: Observing Planetesimal Collisions in the Beta Pictoris Debris Disk

Observation	Proposal 9563, Observation 2: Beta Pic - SW Roll 2									Fri Apr 03 16:00:09 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Coronagraphic Imaging									
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Beta Pic - SW Roll 2 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(1)	-bet-Pic	RA: 05 47 17.0877 (86.8211988d)		Proper Motion RA: 5.16 mas/yr					
			Dec: -51 03 59.44 (-51.06651d)		Proper Motion Dec: 84.041 mas/yr					
			Equinox: J2000		Parallax: 0.0509307"					
					Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=Calibration									
	Description=[Coronagraphic]									
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	F210M	BRIGHT (ND Square)	RAPID	9	1	1	1.825	25161
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern	
	A		MASK210R		false		SUB640A210R		NONE	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F182M	F250M	RAPID	4	90	1	90	1885.471	
PSF References	Alpha Pic - SW PSF (Obs 3) (PSF Reference; Filters [F182M/F250M])									
	Additional Justification: false									

Proposal 9563 - Observation 2 - Don't Let the Dust Settle: Observing Planetesimal Collisions in the Beta Pictoris Debris Disk

Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible Aperture PA Offset 2 from 1 by 10 to 14 Degrees (Same offsets in V3)
----------------------	--

Proposal 9563 - Observation 3 - Don't Let the Dust Settle: Observing Planetesimal Collisions in the Beta Pictoris Debris Disk

Observation	Proposal 9563, Observation 3: Alpha Pic - SW PSF										Fri Apr 03 16:00:09 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Coronagraphic Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	-alf-Pic	RA: 06 48 11.4551 (102.0477296d) Dec: -61 56 29.00 (-61.94139d) Equinox: J2000			Proper Motion RA: -66.07 mas/yr Proper Motion Dec: 242.97 mas/yr Parallax: 0.033780000000000004" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Calibration Description=[Coronagraphic]										
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	F210M	BRIGHT (ND Square)	RAPID	5	1	1	1.097	25161	
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern		
	A		MASK210R		false		SUB640A210R		5-POINT-BOX		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F182M	F250M	RAPID	4	10	5	50	1047.484		
PSF References	PSF Reference: true										

Proposal 9563 - Observation 3 - Don't Let the Dust Settle: Observing Planetesimal Collisions in the Beta Pictoris Debris Disk

Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible
----------------------	--